

# Supporting Neurodivergent Students in SLP Master's Programs

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**connect**

# Disclosures

- Reed and Julie are both faculty members at DeSales University.
- Reed is neurodivergent. His teaching philosophy is influenced by his identity and informed by his experiences.

# Learning outcomes

**Note:** This session is intended for faculty, instructors, and clinical supervisors of graduate students, or for individuals wishing to learn more about preparing graduate students.

Attendees will be able to:

- Identify strengths of the neurodiversity paradigm, including the rationale for supporting neurodivergent students in higher education.
- Describe common characteristics and learning needs of neurodivergent students.
- Implement strategies and frameworks to support academic-clinical outcomes of neurodivergent students.

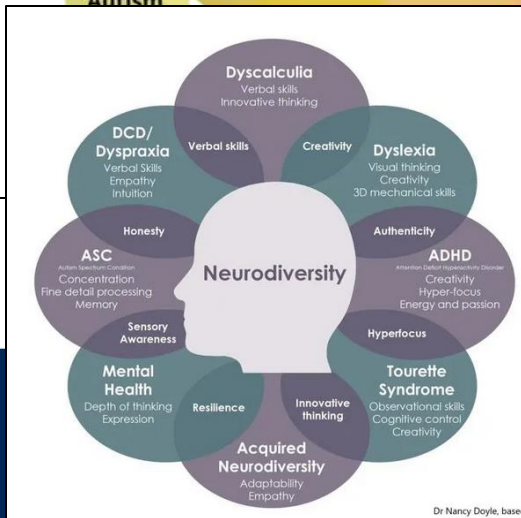
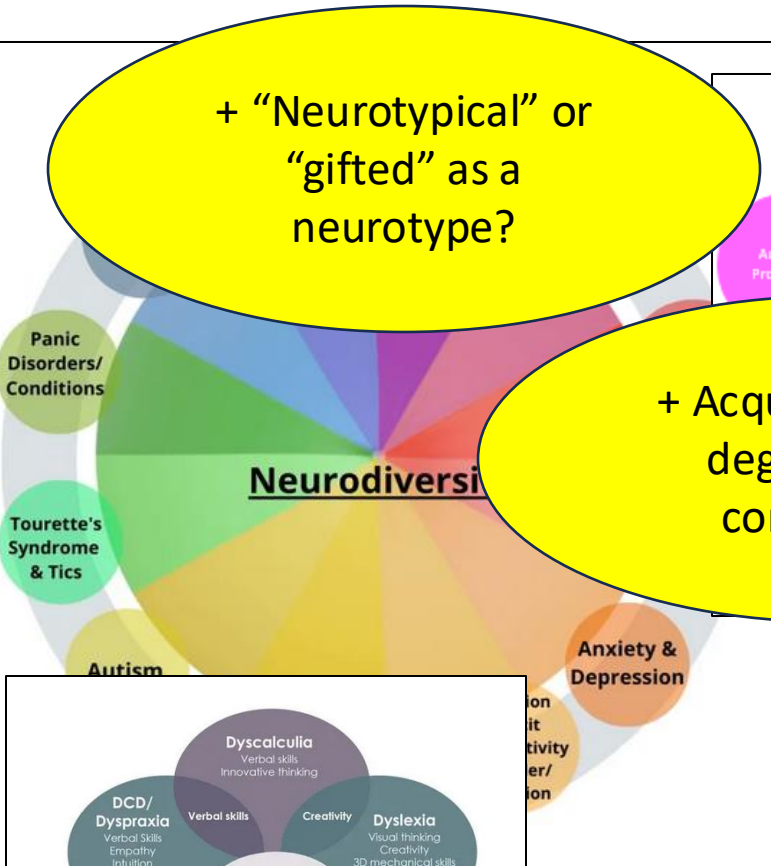
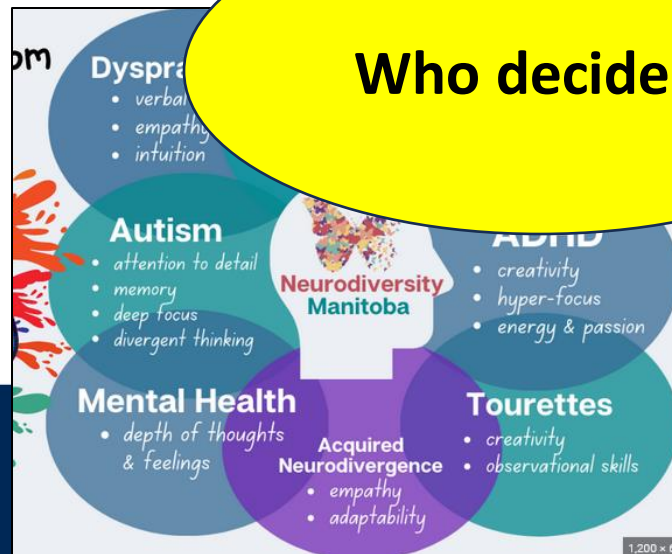
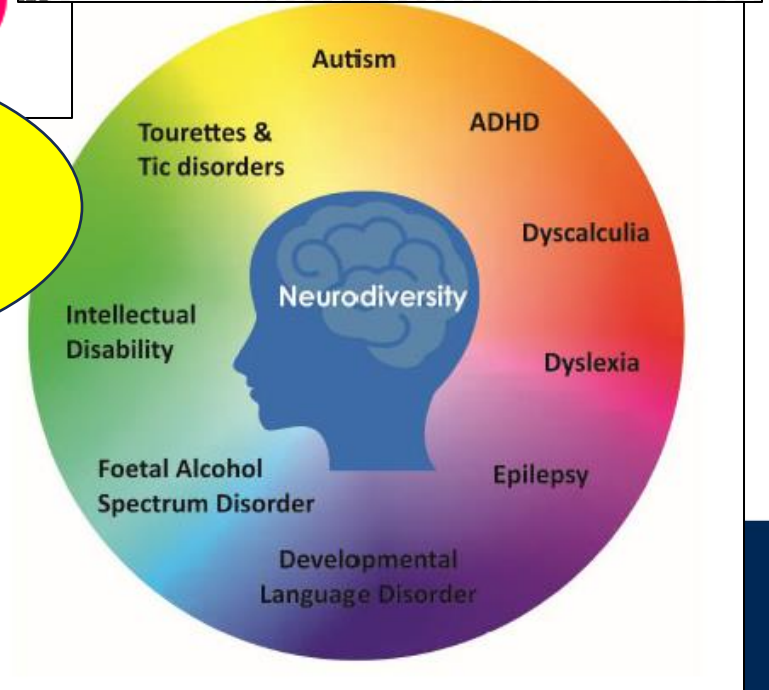
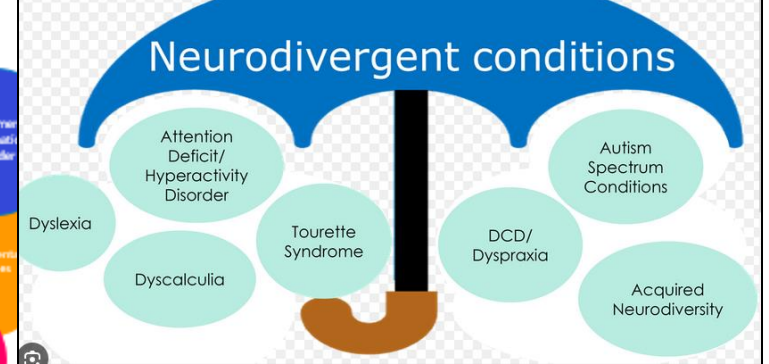
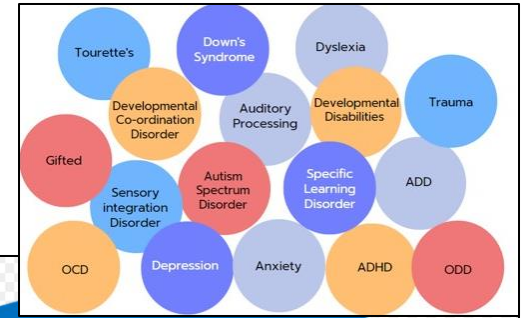
The background is a solid red color. Overlaid on this are several large, semi-transparent red shapes. A prominent feature is a large 'X' shape formed by two diagonal bands crossing in the center. Additionally, there are two large, rounded rectangular shapes, one on the left and one on the right, which are also semi-transparent red. The text 'What *is* neurodiversity?' is centered horizontally and vertically in a white serif font.

What *is* neurodiversity?

# What *is* neurodiversity?

- = “Intrinsic diversity in human brain function and cognition”
  - Biological fact, not opinion!
- May or may not refer to disabilities
  - Clinical significance: Characteristics of neurotype vs. the functional impact
  - “Disability” as a social construct—how does one perceive their neurotype?

# Neurodiversity = Umbrella term



Dr Nancy Doyle, based



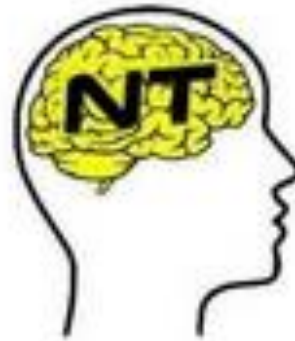
# Terminology

## **NEURODIVERGENT v NEURODIVERSE**



**Each of these people is  
NEURODIVERGENT**

**(There are lots of ways to be neurodivergent)**



**This person is  
NEUROTYPICAL**

**Only a group  
can be  
NEURODIVERSE**

**An individual  
can not.**

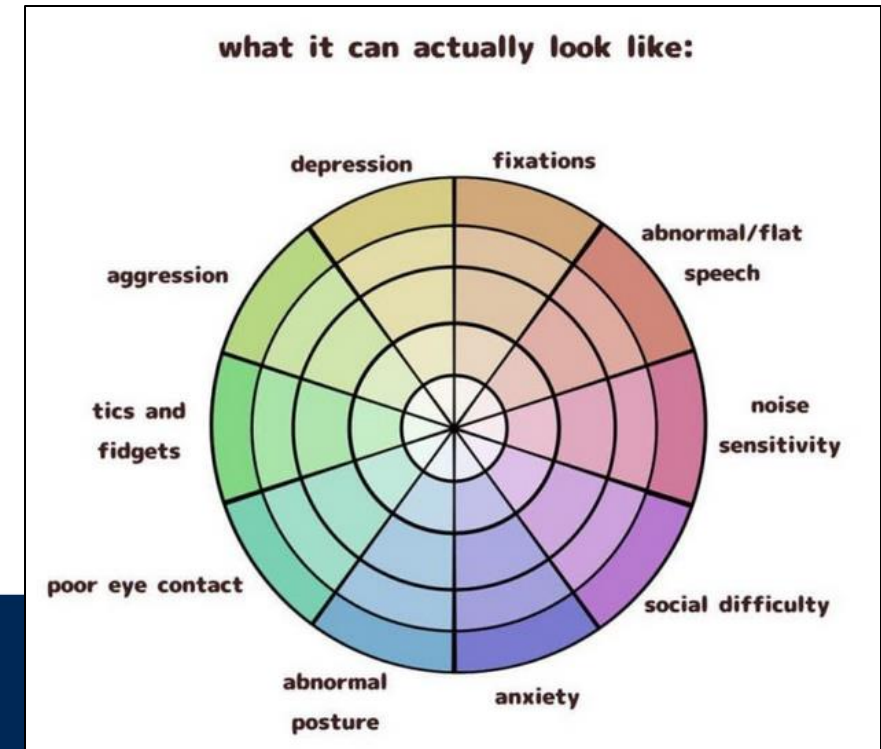
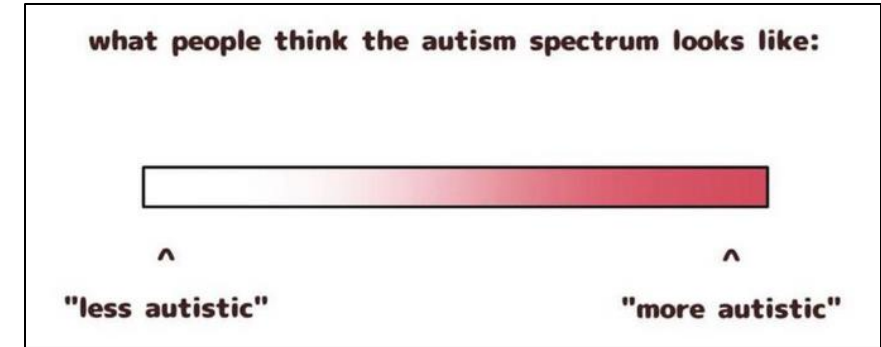
# What *is* neurodiversity?

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- May or may not refer to disabilities
  - Clinical significance: Characteristics of neurotype vs. the functional impact
  - “Disability” as a social construct—how does one perceive their neurotype?
- ...But also, neurodiversity is a:
  - Paradigm
  - Socio-political movement
  - Identity & community



# Variations of neurodivergence

- Considerable within-group variation
  - E.g., “If you know one person with autism, you know *one person* with autism.”
- Considerable intra-person variation
  - Because neurodivergent characteristics aren’t static!
- Considerable between-group overlap
  - Hence myths like “isn’t everyone a little bit autistic?”
  - Or, “That’s not unique to autism, *everyone* does that sometimes.”



# Why do we *encourage* neurodiversity?

Particularly in CSD professions?

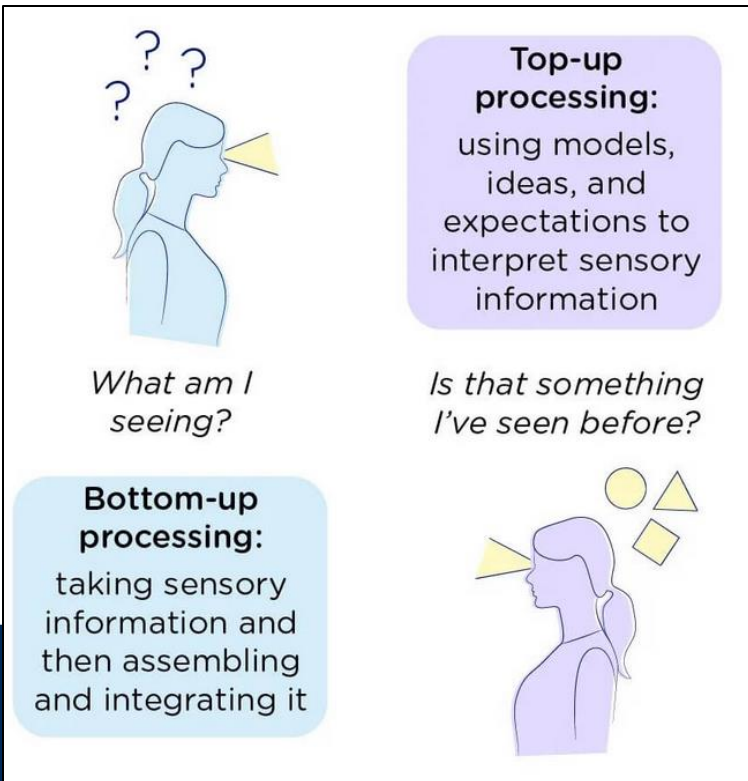
# Strengths of neurodiversity

## 1. Different sets of strengths and weaknesses

- “Top-down” processing vs. “Bottom-up” processing

- Pattern recognition
- Attention to detail

**Do you see  
the forest or  
the trees?**



# Strengths of neurodiversity

## 1. Different sets of strengths and weaknesses

- “Top-down” processing vs. “Bottom-up” processing

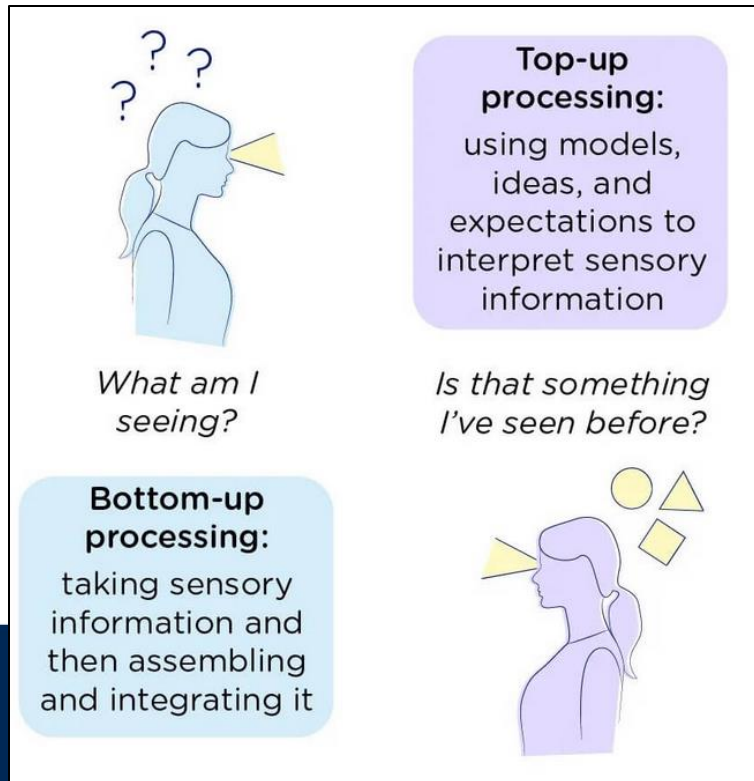
- Pattern recognition
- Attention to detail

- Executive function differences

- Hyperfocus?

- “Spiky” profiles of strengths and weaknesses

- E.g., hyperlexia



# Strengths of neurodiversity

## 2. Uniquely equipped to support neurodivergent clients

If factors such as race and gender impact healthcare relationships and outcomes...

### Implicit Racial/Ethnic Bias Among Health Care Professionals and Its Influence on Health Care Outcomes: A Systematic Review

William J. Hall PhD, Mimi V. Chapman PhD, Kent M. Lee MS, Yesenia M. Merino MPH, Tainayah W. Thomas MPH, B. Keith Payne PhD, Eugenia Eng ... (show all authors)

[+] Author affiliations, information, and correspondence details

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Preference for Same-Race Health Care Providers and Perceptions of Interpersonal Discrimination in Health Care

[Jennifer Malat](#) and [Mary Ann Hamilton](#) [View all authors and affiliations](#)

[Volume 47, Issue 2](#) | <https://doi.org/10.1177/002214650604700206>

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### “It’s Important to Work with People that Look Like Me”: Black Patients’ Preferences for Patient–Provider Race Concordance

Published: 07 November 2022

Volume 10, pages 2552–2564, (2023) [Cite this article](#)

[Carrington Moore](#) [✉](#), [Erica Coates](#), [Ar'Leon Watson](#), [Rebecca de Heer](#), [Alison McLeod](#) & [Arielle Prudhomme](#)

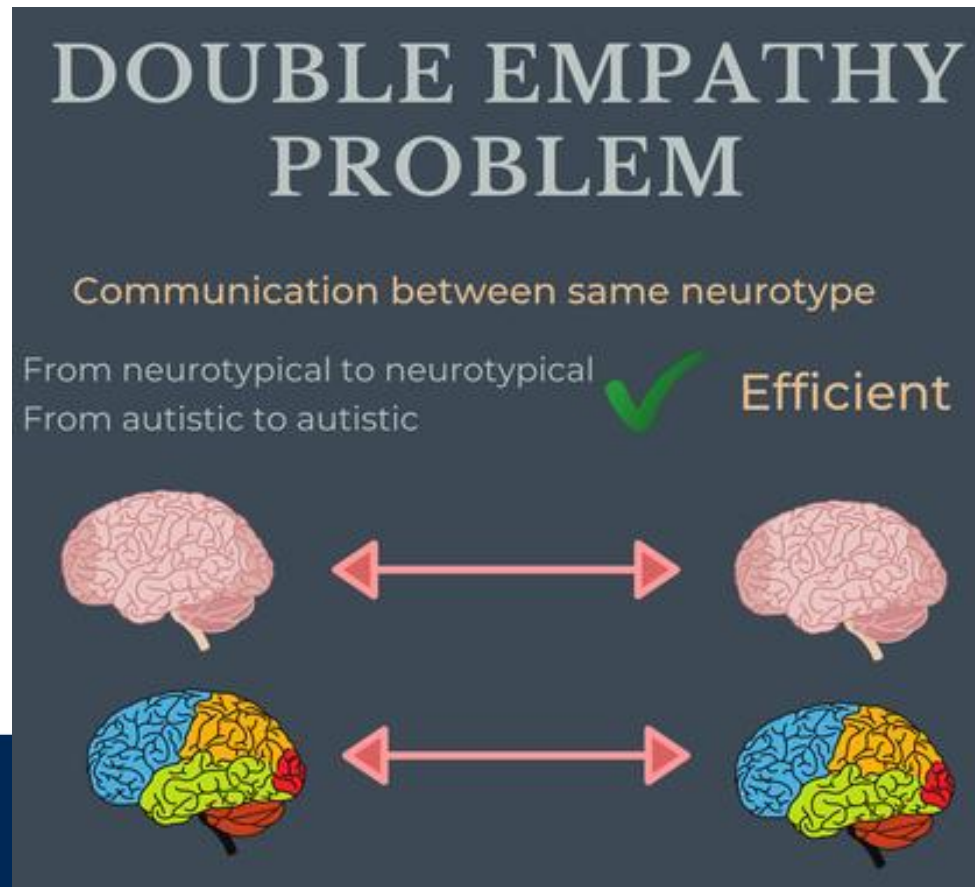
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# Strengths of neurodiversity

## 2. Uniquely equipped to support neurodivergent clients

Then doesn't it make sense if *neurotype* does, too?



There's power when we **connect**

# What challenges might we encounter?

And how can we support our students through  
these challenges?



# Difficulty with content

1. Bottom-up processing
  - Can't see the “big picture”
  - Difficulty seeing how information relates
  - Poor generalization of skills/knowledge
2. Black-and-white thinking
  - Difficulty resolving contradictory (or seemingly-contradictory) info

## Support with:

### Formative assessment

*E.g., implementing low-stakes or no-stakes assignments to evaluate student learning  
(before high-stakes exams or projects)*

## Support with:

### Helping to organize information

*E.g., a blank outline of the lecture (filling in headings, sub-headings, etc.)*

## Support with:

### Explain your rationale

*Understanding why we do things can help students understand when & where to use these skills, or why others might not do these things*

# Difficulty with clinic

## 3. Putting knowledge into practice

### Support with:

#### **Using common terminology**

*Alignment between the language  
& terminology used in the  
classroom & clinic*

### Support with:

#### **Functional, meaningful assignments**

*E.g., assessment “fidelity checks”;  
Coordination between didactic &  
clinical instruction*

### Support with:

#### **Active learning**

*Balancing lecture with in-class  
practice, demonstrations/modeling,  
& assignments*

# Difficulty with executive functioning

## 4. Includes:

- Organization
- Deadlines, procrastination
- Multi-tasking

### Support with:

#### **Scaffolded assignments/task analysis**

*E.g., complete an outline in class before submit a rough draft;  
Breaking a major assignment into smaller components*

### Support with:

#### **Written to-do lists**

*E.g., adding assignment deadlines to students' LMS calendars; listing steps in a multi-step assignment*

### Support with:

#### **Classroom routines**

*E.g., keeping information in the same location every week;  
Requiring regular assignments and/or check-ins*

# Interactions with peers, supervisors, & instructors

## 5. Communication differences and social challenges

- Social nuance (formal, informal)
- Tone
- Directness/bluntness
- Communication/listening “behaviors”
  - E.g., eye contact

### Consider:

**How the student talks to others**

*May benefit from direct instruction & feedback on communication;  
Modeling, scripts, or simulation of difficult conversations*

### Consider:

**How you communicate with the student**

*Clear, direct, and unambiguous communication;  
Feedback should be specific, ideally actionable*

### Consider:

**Prioritize effective communication**

*Focus more on resolving communication breakdowns;  
Reflect on whether something is a difference or a problem*

How can we promote  
inclusive education in  
grad programs?

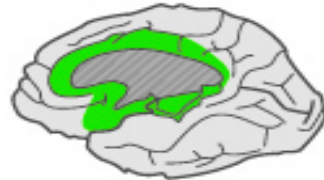
# Universal design for learning (UDL)

- Incorporate multiple means of...
  - Engagement
  - Representation (*i.e., instruction of content*)
  - Action & Expression (*i.e., assessment*)
- Great for all students, not just neurodivergent!
- <https://udlguidelines.cast.org/>



# UDL

Meyer, A., Rose, D., & Gordon, D. (2014). Universal Design for Learning: Theory and Practice. Wakefield, MA: CAST Professional Publishing.



## Provide Multiple Means of Engagement

*Purposeful, motivated learners*

### Provide options for self-regulation

- + Promote expectations and beliefs that optimize motivation
- + Facilitate personal coping skills and strategies
- + Develop self-assessment and reflection

### Provide options for sustaining effort and persistence

- + Heighten salience of goals and objectives
- + Vary demands and resources to optimize challenge
- + Foster collaboration and community
- + Increase mastery-oriented feedback

### Provide options for recruiting interest

- + Optimize individual choice and autonomy
- + Optimize relevance, value, and authenticity
- + Minimize threats and distractions



## Provide Multiple Means of Representation

*Resourceful, knowledgeable learners*

### Provide options for comprehension

- + Activate or supply background knowledge
- + Highlight patterns, critical features, big ideas, and relationships
- + Guide information processing, visualization, and manipulation
- + Maximize transfer and generalization

### Provide options for language, mathematical expressions, and symbols

- + Clarify vocabulary and symbols
- + Clarify syntax and structure
- + Support decoding text, mathematical notation, and symbols
- + Promote understanding across languages
- + Illustrate through multiple media

### Provide options for perception

- + Offer ways of customizing the display of information
- + Offer alternatives for auditory information
- + Offer alternatives for visual information



## Provide Multiple Means of Action & Expression

*Strategic, goal-directed learners*

### Provide options for executive functions

- + Guide appropriate goal-setting
- + Support planning and strategy development
- + Enhance capacity for monitoring progress

### Provide options for expression and communication

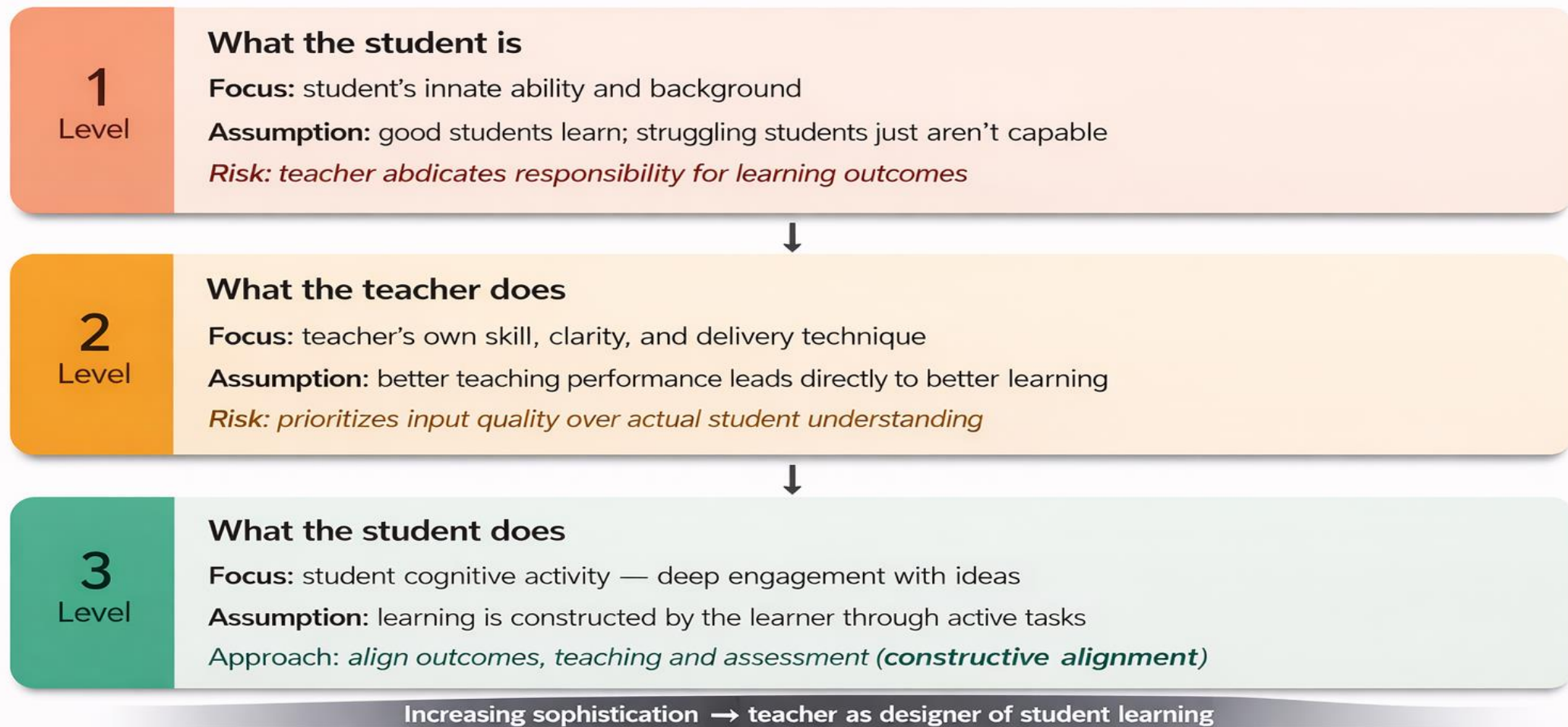
- + Use multiple media for communication
- + Use multiple tools for construction and composition
- + Build fluencies with graduated levels of support for practice and performance

### Provide options for physical action

- + Vary the methods for response and navigation
- + Optimize access to tools and assistive technologies

Figure 1

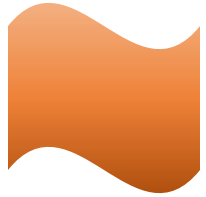
## Biggs' Three Levels of Teacher and Student Learning



Note. Adapted from *Teaching for Quality Learning at University* (4th ed.), by J. B. Biggs and C. Tang, 2011, Open University Press/McGraw-Hill. Copyright 2011 by Open University Press.

Reference Biggs, J. B., & Tang, C. (2011). *Teaching for quality learning at university* (4th ed.), Open University Press/McGraw-Hill.

# Supporting Struggling Students in CSD/SLP: A Case Study Example



## Red Flags/Concerns

- Missed deadlines
- Missing assignments
- Tardiness
- Leaving class frequently
- Superficial/Gist responses
- Talking during class
- Appearing distracted
- Lack of initiation
- Poor communication about struggles/Lack of awareness
- Difficulty task analyzing



## Observable Consequences/Struggles

- Poor prep for assignments and exams
- Poor time management of workload
- Poor performance on academic tasks
- Not meeting competencies – lack of skill development
- Clinical independence –not meeting expected level
- Negative perceptions from faculty and possibly peers
- Not meeting program progression standards



## DSU Program & Faculty Response

- Course remediation with guidance, support, & accountability from faculty
- Probation plan to include regular contact, scaffolding, and accountability
- Clear expectations about initiating and maintaining communication
- EF coaching to support specific areas of need interfering with performance

# Case Study Example: Progress Update

Met course competencies through remediation plans

Improved GPA to meet progression standards (and continues to meet them)

Self advocated and followed through with 1) testing to identify neurodevelopmental differences and 2) seeking accommodations through our Office of Student Accessibility

Continues to implement EF strategies (some not all 😊) in current course work and clinic to maintain organization and time management.

Seeks support and counsel independently when challenged.

# Final considerations

1. Are we setting students up for success?
  - Have they mastered the skills & knowledge necessary to practice?
  - Can they complete the essential functions of an SLP?
    - *Consider access to “reasonable accommodations!”*
2. Where do we draw the line?
  - Traditional perspective: Teach students to conform.
  - ND-affirming perspective: Embrace differences, reject “norms.”
  - Your perspective: *Somewhere in between?*



# Questions / Discussion

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